



Agile teams as complex adaptive systems (CAS)

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Abstract Self-Organizing teams have been studied in various forms and they are considered as an autonomous team operating with minimal dependencies. With the advent of agile methodologies in the late 1990s and the early 2000s, the focus shifted to the team attributes that lead to successful project delivery as compared to the individual attributes only. Agile methodologies are considered as another alternative process as compared to traditional software engineering practices for developing software products and systems. The focus on emphasizing and integrating people and team factors into the software development process forms a core part of agile development. This paper investigates the contemplation of the agile team as a complex adaptive system (CAS) for improved probability of successful project delivery. This is discussed as an important attribute while identifying the key characteristics of agile teams as part of an overall framework of the important characteristics of an agile team that leads to successful project delivery.

Keywords Agile methodologies · Self-organizing teams · Agile teams · Software development · Agile · Complex adaptive systems (CAS)

1 Introduction

With the rise of agile methods in the late 1990s and early 2000s, they provide various options as compared to the long established software engineering processes for developing software products and systems. Agile is focused on social change in the workplace [8, 49]. Agile methodologies focus on integrating various variables like culture, shared understanding of the vision and other factors while undertaking software development and they focus predominantly on the importance of teams and people in the work place [3]. Alistair Cockburn observes—Software development is not a science and hence the scientific method cannot be applied to it [14]. The strength of a team is linked to the member who is the weakest as the team has to work in unison and the weakest member is the constraint in the team [57].

Self-organizing teams have been considered as enablers of organizational holography in the early 1950s, agents of creation of knowledge and management in the 1980s and as instances of units demonstrating instinctive order in Complex Adaptive Systems (CAS) in the 1990s. With the focus on agile methods, the agile team made up of individuals became the central unit around which work began to evolve. Hence, it became important to focus on what may be the key characteristics of agile teams as part of an overall framework that will lead to successful project delivery. When the teams are viewed in the context of CAS, the traditional command and control mindset will need to undergo a change and the focus is on self-organization and autonomy with purpose.

This paper evaluates the role of complex adaptive system (CAS) as an important attribute of agile teams as part of an overall framework identifying the key characteristics of agile teams which lead to successful project delivery.

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Agile teams are observed as complex adaptive systems (CAS) [5]. A CAS is a vibrant network of numerous agents (individuals/people) constantly acting and responding to each other. The control of the CAS is decentralized, not regulated and it is greatly distributed. System coherence is observed on account of the competitive interactions and mutual cooperation among the agents. Thus, the comprehensive behavior of the system is the outcome of the large number of individual decisions made by all the individual agents at any point in time.

As per systems theory, systems are organized into three types: deterministic which are also known as mindless systems; animated which are also known as unminded systems; and social which are also known as multi-minded systems [2, 24]. When human agents are introduced into a system, then the focus is shifted to human behavior. The key point is to focus on how system and subsystem goal align with each other. This is different from the traditional cause and effect models of human behavior. Focus should also be given to exploring the interactions among the various system levels. Traditional general systems theory [9] is not found to be fully adequate to demonstrate research into the key phenomena occurring in agile teams. Further focus on systems theory indicate that when a human agent is introduced, then the nature of the system is also changed and the way to study that system is also changed. Focus needs to be given on studying additional human traits like pride and other factors. These aspects also need to be studied in more detail in future psychological studies [49].

In the conventional systems theory, there are boundaries which separate the system from its environment. This is due to the vibrant interactions among the social network and the porous nature of the organizational boundaries in reality [44]. The nature of individuals is such that it is not modular unlike mechanical systems which are generally modular. If the process models and the system models do not take into account this aspect of non-modularity, then the outcomes will be very different in reality as compared to theory. There is a fine demarcation between a group which is considered as a collective entity as compared to the individuals within a group [10]. A group wields significant influence on the members in the group. The paper focuses on the concept of agile teams made up of individuals and the whole team follows the agile implementation and the agile practices and the psychological and social/cultural interaction among the team members and other teams. (In this paper, to avoid confusion and for purposes of simplicity, a group is considered as a team, even though strictly speaking, a group and a team are different). A group is generally informal and comprised of members who are focused on solving short term problems and issues whereas a team is more focused on the long term

solutions and it requires more coordination, collaboration and structure [57].

2 High level characteristics of CAS

2.1 Foundations of complex adaptive systems (CAS)

A CAS generally consists of a large number of agents who behave according to a set of rules. The rules require the agents to modify their behavior as per the other agents existing in the system. The agents interact with each other and they adapt to each other. Important concepts of CAS are—emergence, hierarchy, self-organization, robustness, feedback loops, edge of chaos and inter-connected autonomous agents. A software development team is expected to be made up of autonomous team members who have their own schemata (values, beliefs, standards) that are embraced by individuals [55, 56].

3 Objective

The focus of this paper is to highlight the role of agile teams as complex adaptive systems and how the focus on CAS influences the identification of framework parameters that make up the properties and characteristics of agile teams and to expound on the commonalities associated with teams in the area of agile development projects. It is propositioned that agile is a methodology that can facilitate teams to be more responsive to the changing market requirements than traditional software development models.

The problem statement is—How an agile team considered as a CAS forms an important attribute of the characteristics of agile teams when viewed as part of an overall framework for identifying the important characteristics of an agile team that leads to a higher probability of successful project delivery. If the attribute of agile teams as a CAS is not taken into account, then the probability of the team to undertake successful project delivery may not be very high and the overall framework may also not be very successful in identifying the key characteristics of agile teams that lead to a higher probability of successful project delivery. The consideration of the attribute—agile team as a CAS as an important attribute of the characteristics of an agile team that lead to an improved probability of successful project delivery and as part of an overall framework is validated through a pilot survey which focused on this attribute.

4 Literature review

The current world of agile software development focused on software product development and customer delivery assumes that people focus on value, eliminate waste, work faster and produce a better quality product, manage change appropriately and focus on novel techniques of undertaking work. It is also based on the idea that if a group of people are brought together and made to work as a team, then due to the team interactions and the theories behind team behavior, it is expected that they will become effective quickly. Thus, if a simple framework indicating key attributes of agile teams is available and which can be applied for a team that is implementing agile methods, then the probability of a successful project delivery is improved considerably as compared to just performing as an agile team and not focusing on the key characteristics which make up an agile team. This will also impact existing agile teams which do not focus much on the key characteristics of agile teams and keep trying to find out how to improve their project delivery further. The literature review indicates that different studies are available regarding frameworks for identifying the key attributes and characteristics of effective teams that will facilitate them to effectively deliver good quality software products as per the changing market requirements in a short period of time. However, the research studies diverge on the key role of agile teams as CAS and which also form part of an overall framework for identifying the important characteristics of agile teams that will facilitate a higher probability of successful project delivery. Table 1 gives the analysis of previous research carried out, study outcome and gap analysis.

Additional research gives more information on the nature of CAS as an important variable while determining the characteristics of agile teams. Complex adaptive systems (CAS) are a basic unit of analysis in complexity science theory. CAS occur naturally in social systems [28, 45, 53]. They are capable of solving problems innovatively and are able to learn and adapt quickly [15, 16, 25, 33, 39, 42]. Mature social systems are made up of a complex of hierarchically embedded, overlapping and interdependent aggregates or CAS [31, 35]. Emergence refers to a non-linear abruptness that characterizes change in complex systems [37]. It derives from the dissipation of built up tensions [51], sudden mergers (or divergences) of formerly separate CAS [31] or a cascade of changes through network connections [7]. Diversification from the viewpoint of complexity science is defined as increasing internal complexity (number and level of interdependent relationships, number of CAS, heterogeneity of skills and outlooks within CAS and tension) to the point of or exceeding that of the competitors or the environment (i.e., “requisite

variety,” [6] or “requisite complexity,” [41]). Dooley [22] describes a CAS as the summative of interacting agents that “behaves/evolves according to three key principles: order is emergent as compared to predetermined, the system’s history is irreversible, and the system’s future is generally unpredictable.” The randomness nature of CAS points out that complex dynamics can display sensitivity to minor perturbations and the changes are non-linear [36], CAS are rather organic and unpredictable [38]. By increasing their complexity, CAS enhance their ability to process data [34], solve problems [33], learn [15, 33], and change creatively [37].

The above literature study indicates that the focus on CAS in the agile context forms an important part of the characteristics of agile teams.

5 Proposed research model

The Research Model proposed for the overall framework for identifying the important characteristics of agile teams that lead to a higher probability of successful project delivery is given below (Fig. 1).

Specific sample hypotheses were worked out based on the research model and for this paper, the hypothesis for the independent variable—agile team as CAS (X9) was defined and validated as part of the pilot study. Other hypotheses for each of the other variables as per the above model will need additional analysis of samples and validation will subsequently need to be carried out and which is not part of the scope of this paper.

H_a: There is a significant positive relationship between the improved work outcomes for the software project and the application, understanding and consideration of the agile team as a complex adaptive system (CAS).

Explanation of all the variables used in the model is given below:

Selection of Team Skills includes the selection of members who are able to work in a team working in a complex environment and the focus on competency and capability to carry out the specified task.

Behavioral Factors includes maturity, ability to understand complexity and work in a complex environment.

Leadership includes the ability to work in a complex environment and able to take appropriate decisions for complex scenarios.

Motivational Factors include intrinsic motivation, empathy and ability to solve complex scenarios through persistence.

Organizational Culture refers to the culture existing in an organization where the work is being undertaken.

Table 1 Analysis of previous work, study outcome and gap analysis

Sl. No.	Previous work area	Authors	Study outcome with gap analysis
1	Focus on agile teams approaching uncertainty as threats and opportunities	Dönmez and Grote [21]	Approaches to teams dealing with different types of uncertainties Not much focus on other characteristics of agile teams including CAS as part of a framework
2	Focus on understanding complex adaptive behavior of hybrid teams	Canana et al. [13]	Focus on problem solving as a team behavior in the complex domain as compared to individual behavior Not much focus on CAS in teams and various attributes of agile teams as part of a framework that leads to successful project delivery
3	Focus on teamwork in agile teams using adapted Big Five teamwork theory	Strode [60]	Focus on teamwork in agile teams Adapted form of Big Five teamwork theory to explain factors affecting team work in agile software development teams Framework does not consider all the factors that may affect the characteristics of agile teams including CAS fully
4	Focus on understanding which are the factors that affect the productivity of agile teams	de Melo et al. [19]	Focus on factors that lead to improved work outcomes of agile teams Development of a framework using thematic analysis Focus on agile team management However, other factors like complex adaptive systems, behavioral factors, disruptive innovative techniques were not fully considered Hence, a comprehensive framework for identifying the characteristics of agile teams that lead to improved work outcomes is not available
5	Team Performance in Agile SW Teams. Focus on attributes that affect performance of the team	Dingsøy and Lindsjörn [20]	Findings from 18 Focus Groups Focus on factors—mutual trust, orientation of the team, adaptability, team leadership, backup behavior, shared mental models, mutual performance monitoring and closed loop communication Big Five teamwork model of Salas Model adopted as the context for the study Does not focus on other factors that may also be important for teamwork like disruptive innovative techniques for problem solving, team as a complex adaptive system (CAS)
6	An investigation of the decision making process in agile teams	Meghann and Orla [43]	Exploration of the decision making process in agile teams Identify factors that influence the decision making process during the sprint planning and daily scrum meetings However, no focus on CAS and an overall framework on the important attributes for agile teams

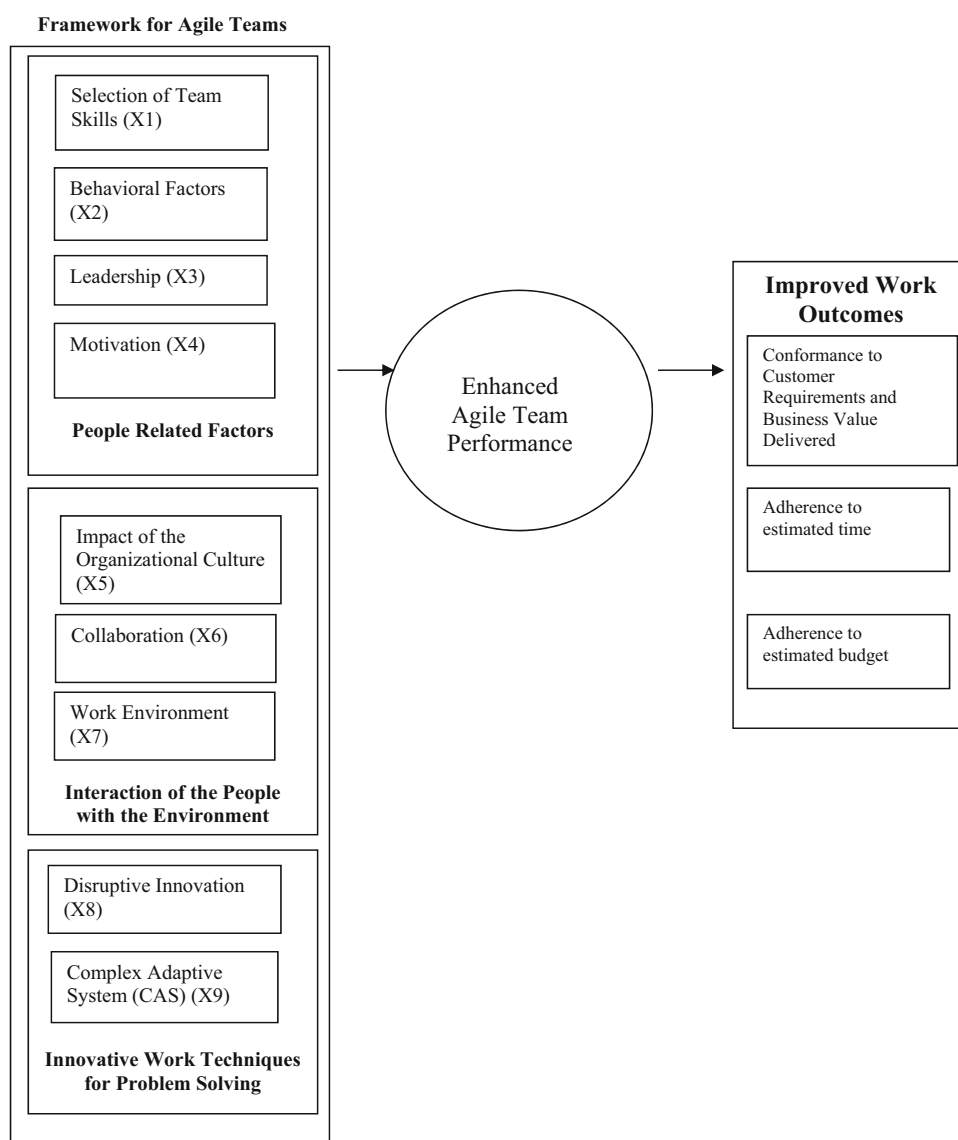
Table 1 continued

Sl. No.	Previous work area	Authors	Study outcome with gap analysis
7	Study focused on self-monitoring SW development team performance using an instrument and analyzing the findings	Kettunen and Moilanen [32]	Focus on an instrument for self-monitoring agile teams and how to improve the performance of the team All the characteristics of agile teams are not fully covered like work environment and other factors that lead to improved work outcomes
8	Agile teams are better suited for rapid and quality delivery as per the changing market requirements	Dyba and Dingsøyr [23]	Discusses the role of agile teams and how they are better suited for managing changing market requirements. However, no specific focus on the complex adaptive system as part of an overall framework is indicated
9	Agile software development encourages evolutionary development and encourages rapid and flexible response to change	Williams and Cockburn [66]	Discusses the key concepts of agile software development and how it harnesses change for managing the customer requirements effectively. However, it does not indicate the key characteristics of agile teams in the context of a complex adaptive system as part of an overall framework
10	Returns derived from vastly effective and focused teams and frameworks for high performing teams	Nedelko [48]; [52]; Ulloa and Adams [63]; Daniel and Davis [18]; Hoegl and Proserpio [26]; Thamhain [62]; Salas et al. [54]	Different studies focus on the advantages of focused teams and the different types of frameworks for high performing teams Focus on CAS as a key variable is not indicated
11	How agile software development methodologies could help facilitate the development of robust software systems for the marketplace	Dyba and Dingsøyr [23]; Abrahamsson et al. [1]	Focus on agile software development methodologies that help facilitate the development of software systems Focus on CAS as a key variable is not indicated
12	Focus on Social nature of agile teams and the explanation of how social identity and collective effort are supported by agile methods	Whitworth and Biddle [65]	Study focuses on the social nature of agile teams and explains social identity and collective effort are supported by agile methods Focus on CAS as a key variable is not indicated
13	Building agile teams focuses on the study of highly accomplished teams	McGeachy [40]	The results of the study are applied to teams that have embraced the Agile methodologies Focus on CAS as a key variable is not indicated
14	Concept of social identity is a very important characteristic for the agile teams	Ilgen et al. [29]	Study pertaining to self-organized work teams has focused on social psychology and social identity Focus on CAS as a key variable is not indicated
15	Individual psychology operating in the social context is focused in the Social Identity theory	Tajfel and Turner [61]	Social perspective is focused to highlight the importance in describing various characteristics of agile teams Focus on CAS as a key variable is not indicated

Collaboration includes communication among the team members apart from how they collaborate with each other to meet the goal.

Work environment includes different types of work environment like physical and virtual work environment.

Disruptive Innovation methods includes capability to solve complex scenarios using appreciative interviews, design story boards and other techniques.

Fig. 1 Research model

Consideration of the agile team as a CAS is the ninth variable in the above framework and this paper focuses only on this variable.

5.1 Model and variable definitions

The variables indicated in the above model are explained as given below -

- (X1, X2, X3, X4) People Related Factors [40, 65]—1, 2, 3, 4.
- (X5, X6, X7) Interaction of the people with the environment [40]—5, 6, 7.
- (X8, X9) Innovative Work Techniques for Problem Solving (McCandless Keith, Lipmanowicz, Henri, 2014; [38]—8, 9.

The following model is used for testing the overall hypotheses:

$$\begin{aligned} \text{Improved Work Outcome of the Project} \\ = \beta_0 + \beta_1 X1 + \beta_2 X2 + \beta_3 X3 + \beta_4 X4 + \beta_5 X5 \\ + \beta_6 X6 + \beta_7 X7 + \beta_8 X8 + \beta_9 X9 + E_i \end{aligned}$$

(where— E_i indicates residuals in the equation).

The above proposed model suggests that multiple linear regression is required to be used in order to understand the inter-relationships between the various variables and the effect of these independent variables on the dependent variables. However, in this paper, as the focus is on CAS, only linear regression is used. Other variables indicated in the model also need to be validated as part of a wider study, but it is not part of the specific scope of this paper.

5.2 Research design

1. Type of research: Descriptive Research and in this paper, the focus on the role of agile team as a CAS is explored.
2. Comparison of proposed technique with techniques used by other researchers—The pilot study used random sample survey and linear regression along with a structured questionnaire to arrive at the findings. Other techniques used by researchers were—Interpretive research approach along with an exploratory research study (semi structured interviews with open ended questions) [64]; Secondary research of data along with cataloguing of the information and discussions on key areas [30]; Discussions approach using Quantum decision theory in Hilbert space [13]. The techniques used in this paper are focused on direct discussions with the respondents from two organizations based on a random sample survey using a structured questionnaire. Subsequently, linear regression was applied on the data to arrive at the findings. Other studies take alternative approaches to expound on their findings. Hence, a comparison of our proposed techniques with techniques used by other researchers underlines our focus on a practical and realistic approach to arrive at the findings.
3. Data collection method: Primary method of data collection is based on discussions/interviews using a structured questionnaire. A simple random survey was used for the pilot study.
4. Sampling: Simple Random sampling since a simple random sample is an unbiased technique of collecting data without bias:
 - a. Frame: IT/ITeS Organizations involved in software development—two organizations for the pilot study.
 - b. Elements: Managers, team leads and software developers, Agile Coach, ScrumMaster.
 - c. Experience level: In Years.
- V. Data analysis: The researchers used statistical tools like linear regression analysis besides descriptive statistics as part of the research design. Analysis of qualitative data is done using Likert Analysis [11]. Fifty samples were considered for the pilot study made up of Managers, team leads and software developers, Agile Coach and ScrumMaster. The pilot sample was spread across two organizations—one in—Bangalore city, Karnataka and the other in Noida city, Uttar Pradesh.

6 Methodology

Based on the identification of the various parameters, it was decided to go ahead with a sample survey that will focus on the agile team as a CAS and to get the feedback from the members of the team and agile coaches, project manager, team lead and other members. The questions focused on areas related to the role of CAS as a key variable in agile teams. The methodology adopted was to initially create a questionnaire based on the independent variable—CAS that was identified and the focus was on a small sample size through a pilot study administered in two software organizations—one in Bangalore, Karnataka and the other in Noida, Uttar Pradesh. The questionnaire was administered and discussions were held with the key respondents, namely—project managers, agile coaches, team leads, software engineers and design engineers. The qualitative data was gleaned through discussions held with the key respondents.

A questionnaire identifying the important framework parameters was prepared. Likert Scale was used for ranking the choices of the members on a scale of 1–7 with 1—Strongly Disagree, 4—Undecided and 7—Strongly Agree being the mid and end choices in the scale and the other numbers being higher or lower as per the choice. The questionnaire targeted both the independent and dependent variables. A sample of fifty respondents was selected and the sample questionnaire was administered to them. Based on the feedback obtained, a linear regression test was conducted.

Additionally, the null and alternative Hypotheses were also formulated for the independent variable and the alternative hypothesis is:

H_a : There is a significant positive relationship between the improved work outcomes for the software project and the application, understanding and consideration of the agile team as a complex adaptive system (CAS). The alternative hypothesis was tested and validated as part of the pilot study.

6.1 Study findings

Linear Regression analysis generates an equation to describe the statistical relationship between one or more predictor variables and the response variable. In statistical modeling, regression analysis is a statistical process for estimating the relationships among variables. It includes many techniques for modeling and analyzing multiple variables, when the focus is on the relationship between a dependent variable and one or more predictor variables.

In statistics, the Durbin–Watson statistic is a test statistic used to detect the presence of autocorrelation (a

relationship between values separated from each other by a given time lag) in the residuals (prediction errors) from a regression analysis. Values approaching 0 indicate positive autocorrelation.

R-squared values range from 0 to 1 and are commonly stated as percentages from 0 to 100%. The below table (Table 2) gives the details of the study findings (sample size—50 respondents).

ANOVA

Model	Sum of Squares	Df	Mean Square	F	Sig.
1 Regression	11.055	1	11.055	74.478	.000 ^a
Residual	7.125	48	.148		
Total	18.180	49			

Dependent Variable: Yvariables

^aPredictors: (Constant), CAS

If the improvement due to fitting the regression model is much greater than the inaccuracy within the model then the value of F will be greater than 1. For the model, the F -ratio is 74.478, which is very unlikely to have happened by chance ($p < .001$). This implies that the model significantly improved the ability to predict the dependent variable.

Coefficients

Model		Unstandardized coefficients		Standardized coefficients	t	Sig.	95.0% Confidence interval for B
		B	Std. Error				Lower bound
1	(Constant)	1.148	0.516		2.224	0.031	0.110
	CAS	.789	.091	.780	8.630	0.000	0.605

Dependent variable: Yvariables

Coefficients

Model	95.0% Confidence interval for B
	Upper Bound
1 (Constant)	2.187
CAS	.972

Dependent variable: Yvariables

The above details indicate the parameters of the model. The B-values tell us about the relationship between the work outcomes and CAS. As the value is positive, we can

infer that there is a positive relationship between CAS and the work outcomes.

Residuals Statistics

	Minimum	Maximum	Mean	Std. deviation	N
Predicted Value	4.3026	6.6682	5.5800	.47499	50
Residual	− 1.09111	.90889	.00000	.38132	50
Std. Predicted Value	− 2.689	2.291	.000	1.000	50
Std. Residual	− 2.832	2.359	.000	.990	50

Dependent variable: Yvariables

The differences between the values of the outcome predicted by the model and the values of the outcome observed in the sample are known as *residuals*. These residuals represent the error present in the model. If a model fits the sample data well then all residuals will be small (if the model was a perfect fit of the sample data—all data points fall on the regression line—then all the residuals would be zero). If a model is a poor fit of the sample data then the residuals will be large.

Standardized residuals with an absolute value greater than 3.29 (we can use 3 as an approximation) are cause for

concern because in an average sample case, a value this high is unlikely to happen by chance. In this case, the standardized residual is less than 3. Hence, the model conforms to a reasonably accurate sample fit.

7 Discussions

Taking into account the key factors in agility and observing their mapping linkages with CAS concepts further strengthens the positive relationship between improved work outcomes and the understanding of the agile team in the context of a complex adaptive system as part of an overall framework identifying the key characteristics of

Table 2 Model Summary [(b) . Dependent Variable: Yvariables]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.780 ^a	.608	.600	.38527	.608	74.478	1
Model	Change Statistics			Durbin-Watson			
	df2			Sig. F Change			
1	48 ^a			.000			
				1.688			

^aPredictors: (Constant), CAS

R gives the values of the multiple correlation coefficient between the predictors and the outcome. The value of *R* being 0.780 indicates that the predictor variable has a significant impact on the outcome of the criterion variable. As $p < 0.05$, it leads to the alternative hypothesis being considered as true and accepted

R^2 is a measure of how much of the variability in the outcome is accounted for by the predictors. In this case, Complex Adaptive System (CAS) being the independent variable accounts for about 60% of the variation in the improved work outcomes (business value, time, cost). Thus, the predictor is a good indicator of the work outcomes and the model is a reasonable fit of the data samples

The adjusted R^2 gives an idea of how well the model generalizes and generally the value is very close to the value of R^2 . In this case, the difference for the final model is small (in fact the difference between the values is $.608 - .600 = .008$ (about 0.8%)). This shrinkage means that if the model were derived from the population rather than a sample, it would account for approximately 0.8% less variance in the outcome

Assessing the accuracy of a model across different samples from the same population is known as cross-validation

Stein's Formula gives an indication of how well the model cross-validates different samples of data from the same population. Stein's formula as given below -

Stein's equation - R^2 is the unadjusted value, n is the number of samples and k is the number of predictors in the model. Using $n=50$ samples and $k=1$, the adjusted R^2 works out to be 0.616 as against 0.608 for observed R^2 which indicates that the cross validity of the model is good

Durbin-Watson Statistic is 1.688. It indicates that the errors in regression are independent; this assumption is likely to be met if the Durbin-Watson statistic is close to 2 (and between 1 and 3). $adjusted R^2 = 1 - \left[\left(\frac{n-1}{n-k-1} \right) \left(\frac{n-2}{n-k-2} \right) \left(\frac{n+1}{n} \right) \right] (1 + R^2)$

agile teams that lead to a higher probability of improved work outcomes. The pilot study indicates that the key facets of agility—autonomous and sharing team when viewed through the CAS lens pertaining to the attribute—inter connected autonomous agents and self-organizing teams indicates a strong linkage [4, 17, 46]. Similarly, from the agility perspective – stability with uncertainty linked to the edge of chaos (CAS concept) [12, 58] is also linked and the third agility variable—team learning is linked to the CAS concept of emergence [47, 58].

The edge of chaos provides organizations with adequate incentive and autonomy to experiment and adjust and also provides appropriate frameworks and structure to ensure that they avoid complete disordered disintegration. Additionally, in order to compete at the edge of chaos, organizations must know what to structure and what not to structure in order to promote communication and to capture cross-business synergies [12]. The edge of chaos concept indicates that being agile is neither chaotic nor static and it needs stability but not to the extent that order is present and innovation is suppressed. It is a subtle balance of both the factors. The concept of emergence focuses on learning which is considered as a collective behavior of producing new thought patterns occurring at the team level based on the interaction of the agents within the team instead of

being seen exclusively as the prerogative of individual learning. Learning focuses not only on training and acquisition of skills but it also focuses on gaining insight and new understanding which leads to original knowledge and behavior. When learning leads to novel behaviors, the team is considered to have adapted and evolved to a new state [46]. An agile team leads to new team learning and creation of new knowledge which is shared among other agents within the team to further generate new learnings, insights, knowledge, wisdom and behavior.

In short, this study focuses on the role of CAS as a key independent variable as part of an overall framework containing various key variables identifying the key characteristics of an agile team that lead to a higher probability of improved work outcomes and focuses on key facets—autonomous and sharing team through agents, team learning and stability and embracing uncertainty. These aspects can be focused only through the consideration of the agile team as a CAS. If we do not consider the agile team as a CAS, then the focus on the CAS attributes are negated and the agile team may not perform very effectively and the work outcomes will also not be very effective. The pilot study also indicates a similar finding. As per the pilot study administered for fifty samples and the application of the linear regression test, the p value is less than 0.05. This

indicates that the alternative hypothesis is validated and considered as accepted. Hence, the pilot study validated the role of CAS as a key independent variable in agile teams as part of an overall framework of various key variables identifying the key characteristics of an agile team that lead to a higher probability of improved work outcomes and if the agile teams are considered in the context of CAS, the role of the other variables making up the characteristics of agile teams as part of an overall framework can be represented and understood in a much better manner and which can lead to a higher probability of project success. It also indicates how the role of CAS influences the inter-relationships among the various variables and the role of other independent variables that constitute the important characteristics of agile teams as part of the overall framework identifying the key characteristics of agile teams that lead to a higher probability of project success.

The principal challenges that were observed are:

Additional Research may be needed to identify the impact of the role of all the other variables making up the important characteristics of agile teams as part of the overall framework and the additional interactions and inter-relationships that may need to be observed to come up with the important characteristics of agile teams that lead to a higher probability of project success. The paper focuses on agile teams and does not focus on non-agile teams even though a specific team has its own definite team characteristics. Thus, additional research may be needed to understand the specific components/characteristics of agile teams that make them truly agile. As part of the changed global scenario, many of the software development activities happen across various time zones and in various countries. This may also impact the software development activities. The paper does not focus on this aspect fully and more data is needed to understand the characteristics of agile teams that may need to be preserved during distributed or global software development (GSD). The paper highlights the role of CAS that needs to be focused when agile teams are set up in an organization. However, the areas highlighted are not exhaustive and detailed research may be needed to explore other elements of the various variables that may also be needed to further explain the key characteristics and attributes of agile teams. Despite extensive interest regarding research on teams, a clear definition of teamwork continues to be weak [54]. There are also many divergent views and related research on what makes a team effective. Additionally, there are numerous sources of guidance regarding how to make the most of a team but in all these cases, there are no easy rules or procedures. A team is made up of distinct and independent individuals and it cannot be generalized easily regarding how these individuals will interact and how the team will continue to make improvements [27]. Team Dynamics is

an important aspect that needs to be considered for effective teams. The concept of a team as a complex adaptive system (CAS) is helpful to understand holistically that the work output of the team is more than the sum of the individual work outputs of the members and the inter-relationships of the various components in the framework may need to be viewed holistically within the context of complex adaptive system theory. However, if the teams can imbibe the appropriate practices, then the teams can hope to achieve high performance [59] and can also become effective agile teams. This paper also does not extensively highlight the specific individual transactions that may need to be present for an agile team to work effectively. Additional work is currently being undertaken to come up with an overall framework validating the key characteristics of agile teams that will lead to a higher probability of successful project delivery (this also includes one characteristic feature identified in this study – agile teams as complex adaptive systems (CAS)).

8 Conclusions

In this paper, the role of CAS as a key variable in agile teams as part of an overall framework of other variables that lead to a higher probability of improved project delivery is discussed and a pilot study conducted to test this aspect has proved to be positive and it is observed that CAS acts as a key variable in agile teams and the characteristics of agile teams need to be studied in the context of CAS and complexity theory. This paper does not extensively highlight the specific individual transactions that may need to be present for an agile team to work effectively and it also does not highlight the impact of the other variables (as part of the framework) on the probability of improved project outcomes. It is observed that if the attributes of an agile team as indicated by the role of CAS as a key variable are present, then there is a greater possibility for the team to succeed in the work place to deliver successful software products and outcomes as per the changing market requirements. However, more detailed studies are required to further substantiate the initial findings based on the literature research, secondary sources of information and pilot study. This study advocates that CAS acts as a profound force which is at work in agile teams and which highlights the value of agile methodologies. Agile practices such as daily standups are seen to provide high levels of social acceptance, accountability and support, information radiators which improve the team awareness regarding the activities being undertaken by the project team, cross pollination of information and other factors is seen to enhance the emotional state of the individuals toward safety, security and control in the work environment.

Investigation and consideration of CAS as a key variable in software development teams is essential as the software engineering area focuses on agile teams which are self-organized teams. Thus, the paper aims at highlighting the role of CAS as a key variable while considering the characteristics of agile teams as part of an overall framework that leads to a higher probability of improved project outcomes and if this aspect is considered, then the probability for the team to work as a high performance agile team and deliver successful software products in the market place is improved considerably.

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